

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1716

Simon Street 138 kV
February 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1716. The Interconnection Customer proposes a 256.5 MW solar photovoltaic facility interconnecting at the Simon Street 138 kV substation in the Pennsylvania area. The request was received on April 11, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 427 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,820,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Mighoward / Simon Street 138kV line, the Roseville / Dug Gap 138kV line loads to 144.5% of its 177 MVA rating. Project DFAX: 58.9%.

For the loss of Mighoward / Simon St 138kV, the Dug Gap / George 138kV loads to 132.3% of its 199 MVA rating. Project DFAX: 58.9%.

The Slaer / George 138kV L is overloaded to 129.4% of its 211 MVA emergency rating for the loss of the Mighoward / Simon St 138kV. The project contributes 58.9% DFAX (151.2 MW) to this overload.

The N Market / Roseville 138kV L is overloaded to 122.4% of its 197 MVA emergency rating for the loss of the Simon St / Mighoward 138kV line. The project contributes 58.9% DFAX (151.2 MW) to this overload.

The Mighoward / Simon St 138kV is overloaded to 118.4% of its 242 MVA emergency rating for the loss of the George / Sater 138kV. The project contributes 78.4% DFAX (201.1 MW) to this overload.

Foster Wheeler Mt Carmel Cogen / Mighoward 138kV: 109.4% loading (250 MVA) following the outage of the George / Slater Substation 138kV; DFAX = 78.4%.

N Market / Frackville 138kV L: 105.6% loading (220 MVA) following the outage of the Mighoward / Simon St 138kV; DFAX = 58.9%.

Simon Street / Slater Substation 138kV: 104.5% loading (239 MVA) following the outage of the Smon St / Mighoward Substation 138kV; DFAX = 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Roseville / Dug Gap 138kV Line	Rebuild line with higher-capacity conductor	\$3,864,000	100% (\$3,864,000)
Dug Gap / George 138kV	Rebuild line with higher-capacity conductor	\$3,464,000	100% (\$3,464,000)
Slaer / George 138kV L	Rebuild line with higher-capacity conductor	\$2,789,000	100% (\$2,789,000)
N Market / Roseville 138kV L	Rebuild line with higher-capacity conductor	\$3,746,000	100% (\$3,746,000)
Mighoward / Simon Street Substation 138kV	Rebuild line with higher-capacity conductor	\$6,071,000	100% (\$6,071,000)
Foster Wheeler Mt Carmel Cogen / Mighoward 138kV	Reconductor line	\$806,000	100% (\$806,000)
N Market / Frackville 138kV L	Reconductor line	\$2,569,000	100% (\$2,569,000)
Simon Street / Slater Substation 138kV	Reconductor line	\$1,511,000	100% (\$1,511,000)
	Total allocated to project		\$24,820,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.